Complete monotonicity of a squared Mittag–Leffler modulus

Research manuscript

October 28, 2026

Abstract

We prove complete monotonicity of $|E_{2a,1}(-t)|^2 + t|E_{2a,1+a}(-t)|^2$ on $(0,\infty)$ for $0 < a \le 1/2$, the assertion of Conjecture 3 in the preprint of Chorfi–Et-tahri–Maniar–Yamamoto. The proof identifies a positive one-dimensional Laplace representation for the half-order cross kernel, then transfers it to arbitrary a through finite normalized Beta averages and Euler's limit for the Gamma function. Classical integral and convexity tools are used; no new general closure principle is claimed.

1 Statement and the half-order kernel

For $a, b > 0$, let $E_{a,b}(z) = \sum_{n\ge 0} z^n/\Gamma(an+b)$.

Theorem 1. *For $0 < a \le 1/2$, the function*

$$f_a(t) = |E_{2a,1}(-t)|^2 + t|E_{2a,1+a}(-t)|^2$$

is completely monotone: it is smooth on $(0,\infty)$ and $(-1)^k f_a^{(k)}(t) \ge 0$ for every integer $k \ge 0$ and $t > 0$.

The theorem is precisely Conjecture 3, equation (5.2), in [1, Section 5]. The other zero and evolution-equation questions in that source are not addressed here. The genuine even/odd series identity

$$E_{a,1}(z) = E_{2a,1}(z^2) + zE_{2a,1+a}(z^2)$$

and the reality of the two series at negative real arguments give $f_a(t) = |E_{a,1}(i\sqrt{t})|^2$ for $t \ge 0$.

Put $I(x) = \int_0^1 e^{xu^2}\,\mathrm{d}u$. Termwise integration of the exponential series, dominated on $[0,1]$ by $e^{|x|}$, gives

$$\int_0^1 e^{-x(1-u^2)}\,\mathrm{d}u = \frac{\sqrt{\pi}}{2}E_{1,3/2}(-x), \qquad E_{1/2,1}(ir) = e^{-r^2}\left(1 + \frac{2ir}{\sqrt{\pi}}I(r^2)\right). \tag{1}$$

For completeness, the polynomial moments used in the first identity are

$$B_n = \int_0^1 (1-u^2)^n\,\mathrm{d}u = \frac{\sqrt{\pi}\,n!}{2\Gamma(n+3/2)}.$$

Integrating the derivative of $u(1-u^2)^{n+1}$ yields $B_0 = 1$ and $B_{n+1} = 2(n+1)B_n/(2n+3)$; the displayed formula follows from the Gamma recurrence. The second identity in (1) follows from the even/odd decomposition and $E_{1,1}(z) = e^z$.

For $v, w \ge 0$ and $t \ge 0$, define

$$K(v,w,t) = \mathrm{Re}\Big[E_{1/2,1}(i\sqrt{vt})\overline{E_{1/2,1}(i\sqrt{wt})}\Big].$$

Writing $S = v + w$, equation (1) gives

$$K(v,w,t) = e^{-St}\left[1 + \frac{4t\sqrt{vw}}{\pi}I(vt)I(wt)\right]. \tag{2}$$

Lemma 2. *For $v, w > 0$, the expression in (2) equals*

$$\frac{2\sqrt{vw}}{\pi}\left[\int_0^1 \frac{e^{-tw(1-u^2)}}{v+wu^2}\,\mathrm{d}u + \int_0^1 \frac{e^{-tv(1-u^2)}}{w+vu^2}\,\mathrm{d}u\right]. \tag{3}$$

In particular, $K(v, w, \cdot)$ is a positive Laplace mixture. For a zero scale, $K(v, w, t) = e^{-(v+w)t}$.

Proof. Denote the right side of (3) by $J(t)$. Differentiation of ue^{xu^2} and integration give $I(x) + 2xI'(x) = e^x$. The product rule therefore shows that the expression $P(t)$ in (2) satisfies

$$P'(t) + SP(t) = \frac{2\sqrt{vw}}{\pi}\left[e^{-wt}I(wt) + e^{-vt}I(vt)\right].$$

The same equation holds for J: differentiation under its compact integrals cancels each denominator, since $S - w(1-u^2) = v + wu^2$ and $S - v(1-u^2) = w + vu^2$. All integrands and their parameter derivatives are continuous and bounded on compact rectangles. Finally,

$$J(0) = \frac{2}{\pi}\left[\arctan\sqrt{w/v} + \arctan\sqrt{v/w}\right] = 1 = P(0).$$

Thus $e^{St}(P - J)$ has zero derivative and vanishes at zero. This proves the identity. Both weights and both rates in (3) are nonnegative. The zero-scale formula follows directly from (2). □

2 Finite averaging and arbitrary order

Fix $\beta = 2a \in (0, 1]$. For $k \geq 1$, use the normalized positive averaging operator

$$(T_k g)(x) = \beta g(x) + (1-\beta)k\int_0^1 u^{k-1} g(xu)\,\mathrm{d}u.$$

It preserves constants and positivity, and is a contraction in the uniform norm on every interval $[0, C]$. For $q \geq 0$,

$$T_k(x^q) = \frac{k+\beta q}{k+q}x^q.$$

For $N \geq 1$, let A_N be the composition of $T_1, \ldots, T_{N+1}$, evaluated at $C_N = N^{1-\beta}$. Then

$$R_N(q) := A_N(x^q) = N^{(1-\beta)q}\prod_{k=1}^{N+1}\frac{k+\beta q}{k+q} \longrightarrow \frac{\Gamma(1+q)}{\Gamma(1+\beta q)}. \tag{4}$$

This is Euler's Gamma limit. Moreover, $R_N(q)$ increases with N. Indeed, with $d = (1-\beta)q \geq 0$,

$$\frac{R_{N+1}(q)}{R_N(q)} = \left(\frac{N+1}{N}\right)^d \frac{N+2+\beta q}{N+2+q} \geq 1.$$

Here $((N+1)/N)^d \geq 1 + d/(N+1)$ follows from $\log((N+1)/N) \geq 1/(N+1)$ and $e^y \geq 1+y$; multiplying out leaves $d(1+\beta q)/(N+1) \geq 0$. Consequently $0 < R_N(q) \leq \Gamma(1+q)/\Gamma(1+\beta q)$.

Define the genuine complex transforms

$$M_N(z) = A_N\big(E_{1/2,1}(z\sqrt{x})\big) = \sum_{n\geq 0}\frac{R_N(n/2)}{\Gamma(1+n/2)}z^n.$$

The equality follows from uniform convergence on the finite interval $[0, C_N]$. By (4),

$$0 \leq \frac{R_N(n/2)}{\Gamma(1+n/2)} \leq \frac{1}{\Gamma(1+an)}, \qquad M_N(z) \longrightarrow E_{a,1}(z).$$

The entire series on the right supplies a summable majorant on every bounded complex disk, so these are actual transform limits.

Real linearity, complex conjugation and two independent applications of A_N show that

$$|M_N(i\sqrt{t})|^2 = A_N^{(v)} A_N^{(w)} K(v, w, t).$$

This identity is an identity of finite weighted integrals; no existence of a stable distribution is needed. Lemma 2 makes the right side a positive average of exponential Laplace kernels. For arbitrary positive steps $h_1, \ldots, h_k$, every mixed lower difference is nonnegative, because that of $e^{-\ell t}$ is

$$e^{-\ell t} \prod_{j=1}^{k} (1 - e^{-\ell h_j}) \geq 0 \qquad (\ell \geq 0).$$

Finite differences commute with these averages and pass to pointwise limits. Hence all positive mixed lower differences of f_a are nonnegative. The function f_a is smooth by its entire real-series expression. Mixed differences of a smooth function are antitone when one additional lower difference is nonnegative. Taking their negative derivatives and iterating gives $(-1)^k f_a^{(k)} \geq 0$. This proves Theorem 1.

3 Certificate and research status

The Lean endpoint is `MittagModulus.source_conjecture_370772`, a closed proof of the literal two-parameter-series statement. The certificate proves the series convergence, real and complex analyticity, exact finite moments, actual averaging and squared-average identities, the kernel formula including zero scales, and the finite-difference-to-derivative passage. Its pinned environment is Lean 4.34.1 and Mathlib `d13f23b723b`. A fresh isolated source replay passed all 17 modules and 155 declarations, with warnings as errors and only the standard axioms `propext`, `Classical.choice`, and `Quot.sound`. No independent formal-review pass is asserted in this draft.

This manuscript and candidate have zero discovery credit pending source, substantive, prior-art and independent final review, followed by the final packet replay. The preprint's actual conjecture page has been read; the published journal body has not yet been compared. The source is also listed as Chaos Nonlinear Sci. Numer. Simul. **157** (2026), 109758, doi:10.1016/j.cnsns.2026.109758. No worldwide-priority claim is made.

References

[1] S. E. Chorfi, F. Et-tahri, L. Maniar and M. Yamamoto, *Forward and backward problems for abstract time-fractional Schrödinger equations*, arXiv:2510.03600v1 (2025), Section 5, Conjecture 3, equation (5.2). arxiv.org/abs/2510.03600v1.